

Lecture 13

Cramer Rao Lower Bound (CRLB) of variance of an unbiased estimator.

Let $x_1, \dots, x_n$ be a random sample from $f_\theta, \theta \in \Theta$.
 $\delta(x)$ is any unbiased estimator of some $g(\theta)$ of interest, with finite second moment, $E(\delta^2(x)) < \infty$.
 Consider some function, $\psi(x; \theta)$ with $E(\psi(x; \theta))^2 < \infty$.

Then, by the Cauchy-Schwarz inequality,

$$\text{Cov}^2(\delta, \psi) \leq \text{Var}(\delta) \cdot \text{Var}(\psi), \text{ or,}$$

$$\text{Var}(\delta) \geq \frac{\text{Cov}^2(\delta, \psi)}{\text{Var}(\psi)}$$

But this is of no use, unless $\text{Cov}(\delta, \psi)$ does not depend on δ , and ~~the dependence~~ δ is only through its expectation, i.e., $g(\theta)$.

Result A necessary and sufficient condition for $\text{Cov}(\delta, \psi)$ to depend on δ only thro' $E(\delta) = g(\theta)$ is $\text{Cov}(U, \psi) = 0 \forall \theta$, and $\forall U \in \mathcal{U}$, the class of unbiased estimators of 0, i.e., for any $U \in \mathcal{U}$, $E(U) = 0 \forall \theta$. Further $E(U^2) < \infty$.

Pf Sufficiency Assume, $\text{Cov}_\theta(U, \psi) = 0 \forall \theta, \forall U \in \mathcal{U}$ with $E(U^2) < \infty$.
 Let δ_1, δ_2 be any 2 unbiased estimators of $g(\theta)$. Then $\delta_1, \delta_2 \in \mathcal{U}$.
 $\therefore \text{Cov}(\delta_1 - \delta_2, \psi) = 0 \forall \theta \Rightarrow \text{Cov}(\delta_1, \psi) = \text{Cov}(\delta_2, \psi) \forall \theta$
 i.e., $\text{Cov}_\theta(\delta, \psi)$ depends on δ only thro' $E_\theta(\delta) = g(\theta)$
 (and not on any particular choice of δ)

Necessity $\text{Cov}_\theta(\delta, \psi)$ depends on δ only thro' its expectation $E_\theta(\delta) = g(\theta)$
 $\therefore \text{Cov}(\delta, \psi) = \text{Cov}(\delta + U, \psi) \forall \theta, \forall U \in \mathcal{U} \because E(\delta) = E(\delta + U) = g(\theta) \forall \theta$
 $= \text{Cov}(\delta, \psi) + \text{Cov}(U, \psi) \forall \theta, \forall U \in \mathcal{U}$
 $\Rightarrow \text{Cov}(U, \psi) = 0 \forall \theta, \forall U \in \mathcal{U}$.

Use $\psi_\theta(x) = \frac{\partial}{\partial \theta} \ln f_\theta(x)$ in the C-S inequality to obtain the Cramer Rao Lower Bound (CRLB) of $\text{Var}(\delta(x))$.
 We complete the various steps considering a r.v. X from f_θ and consider the case of a random sample of size n .

Steps.

2/13.

$$1. \quad E(\psi_\theta(x)) = \int_{\mathcal{X}} \frac{\partial}{\partial \theta} \ln f_\theta(x) \cdot f_\theta(x) dx = \int_{\mathcal{X}} \frac{\partial}{\partial \theta} f_\theta(x) dx = \frac{\partial}{\partial \theta} \int_{\mathcal{X}} f_\theta(x) dx = 0$$

$$2. \quad \text{Var}(\psi_\theta(x)) = E(\psi_\theta(x)^2) = E\left[\frac{\partial}{\partial \theta} \ln f_\theta(x)\right]^2 \stackrel{?}{=} -E\left[\frac{\partial^2}{\partial \theta^2} \ln f_\theta(x)\right] \quad \text{alternative form}$$

Check this relationship. Note, $\frac{\partial^2}{\partial \theta^2} \ln f_\theta(x) = \frac{\partial}{\partial \theta} \left[\frac{\partial}{\partial \theta} \ln f_\theta(x) \right]$

$$= \frac{\partial}{\partial \theta} \left[\frac{1}{f_\theta(x)} \frac{\partial}{\partial \theta} f_\theta(x) \right]$$

$$= -\left[\frac{1}{f_\theta(x)} \cdot \frac{\partial}{\partial \theta} f_\theta(x) \right]' + \left[\frac{1}{f_\theta(x)} \frac{\partial^2}{\partial \theta^2} f_\theta(x) \right]$$

Take expectations in both sides, use $E[\psi_\theta(x)] = 0 \forall \theta$, and complete it!

$$3. \quad \text{Cov}(\delta, \psi) = E[\delta(x) \psi_\theta(x)] = \int_{\mathcal{X}} \delta(x) \frac{\partial}{\partial \theta} \ln f_\theta(x) f_\theta(x) dx = \int_{\mathcal{X}} \delta(x) \frac{\partial}{\partial \theta} f_\theta(x) dx$$

$$= \frac{\partial}{\partial \theta} \int_{\mathcal{X}} \delta(x) f_\theta(x) dx = \frac{\partial}{\partial \theta} E(\delta(x)) = \frac{\partial}{\partial \theta} g(\theta) = g'(\theta).$$

4. (Verify the NSC)

$$\text{Cov}(U, \psi) = E(U\psi) = \int_{\mathcal{X}} u(x) \frac{\partial}{\partial \theta} \ln f_\theta(x) f_\theta(x) dx = \int_{\mathcal{X}} u(x) \frac{\partial}{\partial \theta} f_\theta(x) dx$$

$$= \frac{\partial}{\partial \theta} \int_{\mathcal{X}} u(x) f_\theta(x) dx = \frac{\partial}{\partial \theta} E(u(x)) = 0$$

$$\text{Hence, } \text{Var}(\delta(x)) \geq \frac{(g'(\theta))^2}{E\left[\frac{\partial}{\partial \theta} \ln f_\theta(x)\right]^2} \quad \left(\text{or } \frac{g'(\theta)}{-E\left[\frac{\partial^2}{\partial \theta^2} \ln f_\theta(x)\right]} \right)$$

R.H.S. is the CRLB of any unb. est. $\delta(x)$ of $g(\theta)$

Note 1. $E\left[\frac{\partial}{\partial \theta} \ln f_\theta(x)\right]^2 = I(\theta)$, is the Fisher information (that the r.v. X contains about the parameter θ)

2. If for some unb. est. $\delta(x)$, $\text{Var}(\delta(x)) = \text{CRLB}$, then $\delta(x)$ is the unique UMVUE of $g(\theta)$.

3. If for some UMVUE, $\delta(x)$, $\text{Var}(\delta(x)) > \text{CRLB}$, then the UMVUE does not attain or achieve its CRLB.

2.4.3 should be considered in the context of a random sample of size n , so that $\delta(\underline{X})$ is the estimator.

Modification in case of random sample, say, $x_1, \dots, x_n$ from

(instead of a $X \sim f_\theta$)

Note there is no change in $g'(\theta)$

only for $I(\theta)$, which is now, $I(\theta) = E \left[\frac{\partial}{\partial \theta} \ln f_\theta(x) \right]^2$ (or, $-E \left[\frac{\partial^2}{\partial \theta^2} \ln f_\theta(x) \right]$)
(say) $= n I(\theta)$ (check this simple modification)

Hence, finally, for a random sample of size n ,

iid from f_θ , $\delta(x)$ any unb. est. of $g(\theta)$, then

$$V(\delta(x)) \geq \frac{(g'(\theta))^2}{n I(\theta)}$$

The CRLB is defined under certain regularity conditions.

1. $\theta \in \Theta$ is an open interval (single parameter)

2. $f_\theta(x)$'s in the family of dists $\mathcal{F} = \{f_\theta(x) : \theta \in \Theta\}$ have a common support, i.e., the set $\mathcal{X} = \{x : f_\theta(x) > 0\}$ does not depend on θ .

3. for any $x \in \mathcal{X} \times \Theta$, $\frac{\partial}{\partial \theta} f_\theta(x)$ exists (and is finite)
(Further, the derivatives mentioned in the defn exist and are finite)

The regularity conditions hold for location-scale families if range does not depend on θ & for exponential families.

Ex 1 $x_1, \dots, x_n$ iid Bin. $(1, \theta)$, $\theta \in (0, 1)$; $g(\theta) = \theta$

$$f_\theta(x) = \theta^x (1-\theta)^{1-x}, \quad x=0,1$$

$$\ln f_\theta(x) = x \ln \theta + (1-x) \ln(1-\theta)$$

$$\frac{\partial}{\partial \theta} \ln f_\theta(x) = \frac{x}{\theta} - \frac{(1-x)}{1-\theta}; \quad \frac{\partial^2}{\partial \theta^2} \ln f_\theta(x) = -\left[\frac{x}{\theta^2} + \frac{(1-x)}{(1-\theta)^2} \right]$$

$$\Rightarrow E \left[\frac{\partial^2}{\partial \theta^2} \ln f_\theta(x) \right] = -\left[\frac{\theta}{\theta^2} + \frac{(1-\theta)}{(1-\theta)^2} \right] \Rightarrow I(\theta) = \frac{1}{\theta(1-\theta)}$$

$$\text{and } \text{CRLB} = \frac{1}{n \cdot \frac{1}{\theta(1-\theta)}} = \frac{\theta(1-\theta)}{n} \quad (\text{which is incidentally the var. of } \bar{X}, \text{ the UMVUE of } g(\theta) = \theta)$$

Ex 2. $x_1, \dots, x_n$ iid $N(\theta, 1)$, $\theta \in \mathbb{R}$; $g(\theta) = \theta$

$$f_\theta(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\theta)^2}, \quad x \in \mathbb{R}; \theta \in \mathbb{R}$$

$$\ln f_\theta(x) = \text{const.} - \frac{1}{2}(x-\theta)^2$$

$$\frac{\partial}{\partial \theta} \ln f_\theta(x) = (x-\theta), \quad E \left(\frac{\partial}{\partial \theta} \ln f_\theta(x) \right)^2 = E(x-\theta)^2 = 1 = I(\theta),$$

so that $\text{CRLB} = \frac{1}{n}$, again the var. of the UMVUE of $g(\theta) = \theta$, is $\bar{X}$.

What if $g(\theta) = \theta^2$?